ARAVIND SANKAR

AI and Robotics Engineer

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PROFESSIONAL SUMMARY

Motivated and innovative Robotics and Artificial Intelligence professional currently pursuing an MSc in Robotics at The University of Manchester. Equipped with a strong foundation in machine learning, control systems, and embedded engineering, with hands-on experience in developing intelligent and autonomous systems. Proficient in Python, C/C++, and PyTorch, with practical expertise in reinforcement learning, robotics simulation, and real-time system design. Experienced in microcontrollers, sensors, and hardware-software integration to build efficient and reliable embedded solutions. Passionate about advancing robotics and automation through cutting-edge technologies.

EDUCATION

MSc Robotics **Sept 2025 - present**

The University of Manchester , United Kingdom

B.Tech in CSE with AI and Robotics **Sept 2021 - May 2025**

Vellore Institute of Technology , India

CGPA : 8.43

WORK EXPERIENCE

Robotics Intern – Vividobots Private Limited **August 2023 – December 2023**

1. Developed **India's first autonomous façade-painting robot**, integrating 3D mapping, ROS-based navigation, and embedded systems,
2. Designed and optimized **SLAM pipelines**, increasing mapping accuracy and reducing navigation errors in real-world environments.
3. Implemented autonomous navigation algorithms using **ROS**, enabling reliable path planning, obstacle avoidance, and real-time localization.
4. Integrated sensors such as **LiDAR, IMU, and depth cameras** for environmental perception and robust state estimation.
5. Engineered embedded solutions using microcontrollers, facilitating seamless communication between hardware and software components.
6. Conducted iterative prototyping, system integration, and real-time testing to enhance performance, reliability, and scalability.
7. Developed simulation models and validated robotic behaviors in controlled environments prior to deployment.
8. Collaborated with cross-functional teams across mechanical, electrical, and software domains to deliver a fully functional autonomous platform.

Core Member – Technocrats Robotics, VIT **2021 – 2023**

1. Engineered autonomous rover subsystems, including camera calibration, RTSP video streaming, and SLAM integration for real-time perception and navigation.Designed and implemented navigation strategies for challenging terrains, improving obstacle detection, clearance, and traversal efficiency.
2. Developed and optimized perception pipelines using computer vision techniques(aruco markers) to enhance environmental awareness and decision-making.

3. Utilized ROS for system integration, sensor fusion, and modular software development, ensuring scalable and robust robotic architectures.
4. Represented the team at the **European Rover Challenge (ERC)** and **Indian Rover Challenge (IRC)**, delivering high-performance robotic solutions under strict timelines.
5. Achieved outstanding results at the Indian Rover Challenge (IRC): **5th Position Internationally (Debut) and 3rd Rank in India**

PROJECTS AND RESEARCH:

Autonomous Nuclear Inspection Robot

1. Developed a **robotic system along with 6DOF robotic arm** for autonomous detection of flames and sparks in nuclear chambers using the **Intel RealSense D435i depth camera**, achieving **95% detection accuracy** in simulated hazardous environments.
2. Implemented **real-time depth and RGB-based computer vision detection** integrated with autonomous arm actuation for targeted repairs.
3. Designed a **CNN-based decision model** to dynamically select between **A*** and **Dijkstra** path planners, improving navigation efficiency by **96%**.
4. Fully integrated **V-SLAM, RTAB-Map, and MoveIt** in Gazebo, performing **50+ successful trials** with high reliability.
5. Optimized the system for **real-time decision-making**, robust navigation, and coordinated operation of perception and manipulation modules.

Hybrid SAC-PID Controller for Autonomous UAV Stabilization

1. Developed a hybrid control system combining **Soft Actor-Critic (SAC)** reinforcement learning with a **PID controller** for autonomous UAV stabilization.
2. Enabled adaptive tuning of PID parameters to enhance stability, trajectory tracking, and robustness in dynamic environments.
3. Implemented and evaluated the model using **Python**, **PyTorch**, and simulation tools.

Fine-Tuned VQA for Technical Diagrams (Qwen-VL + DPO) – Research Project

1. Fine-tuned a multimodal **Vision–Language Model (Qwen-VL)** using **DPO** and **LoRA** on 10,000+ technical diagrams, achieving 95% accuracy in structured visual reasoning tasks.
2. Developed a **Visual Question Answering (VQA)** framework for interpreting complex diagrams, focusing on multimodal alignment, semantic understanding, and compositional reasoning.
3. Built a scalable inference and evaluation pipeline using **Hugging Face,** enabling reproducible experiments and sub-second latency suitable for real-time/edge deployment.
4. Explored cross-modal representation learning and structured visual reasoning, with applications to automated analysis of complex engineering visual data.

SKILLS

- **Robotics & Autonomous Systems :** ROS/ROS2, Gazebo, RViz, MoveIt, URDF/Xacro; SLAM & 3D mapping (RTAB-Map); multi-sensor integration (RGB-D cameras – Intel RealSense D435i, IMUs, Aruco markers); robotic manipulation; embedded/edge systems
- **Machine Learning & Multi-Modal AI :** Deep learning (CNNs), reinforcement learning; Vision–Language Models (Qwen-VL); multimodal learning & fusion; VQA; DPO fine-tuning, LoRA; feature extraction & data analysis for visual/sensor data
- **Embedded systems & Hardware :** Microcontrollers & Embedded programming(Arduino , Raspberry pi , Jetson Nano , FPGA) , Hardware - Software integration , Communication Protocols (UART, I2C, SPI, TCP/IP, RTSP)
- **Programming :** Python, C/C++, JAVA , SQL , application development (Django, PyQt)